

Hemanth Vedantam

AI & Software Engineer

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PROFESSIONAL SUMMARY

AI & Software Engineer with hands-on experience building production-grade AI applications, scalable backend services, and data engineering solutions. Experienced in Python, FastAPI, Agentic AI, LLM applications, PySpark, and SQL through enterprise software development at IKS Health.

TECHNICAL SKILLS

Programming	Python, SQL
Backend Development	FastAPI, Flask, REST APIs, WebSockets
Artificial Intelligence	Large Language Models (LLMs), Agentic AI Systems, LangChain, LangGraph
Data Engineering	PySpark, Google BigQuery, ETL Pipelines
Databases	DuckDB, Qdrant
Cloud & DevOps	AWS, GCP (<i>Basic</i>), Git, CI/CD, Docker (<i>Basic</i>)

WORK EXPERIENCE

Artificial Intelligence Intern 08/2025 – 03/2026
IKS Health, Hyderabad

- Engineered a machine learning-based data integrity solution to detect schema anomalies and data quality issues across 100+ healthcare datasets, each containing approximately 1 million records with diverse data types, improving the reliability of downstream analytics and reporting workflows.
- Developed an agentic AI analytics tool that leveraged LLMs to convert natural language requests into executable Pandas code, automating data exploration workflows and enabling conversational analysis of enterprise datasets.
- Architected a metadata-driven data validation platform using PySpark, Google BigQuery, and 100+ configurable SQL business rules to automate enterprise-scale validation across 100+ live and batch data tables, improving scalability, maintainability, and consistency of data pipelines.

PROJECTS

Autonomous Business Intelligence Platform | FastAPI, LangGraph, DuckDB, Qdrant 🏠 GitHub

- Architected and developed a production-grade Autonomous Business Intelligence platform using FastAPI, LangGraph, DuckDB, Qdrant, and JWT authentication, enabling users to upload structured datasets and obtain executive-level insights through natural language interactions.
- Designed and orchestrated a multi-agent AI workflow comprising specialized agents for query planning, statistical analysis, root cause analysis, forecasting, retrieval-augmented generation (RAG), recommendation generation, and executive report synthesis to automate end-to-end business analytics.
- Engineered scalable backend services for dataset ingestion, metadata management, vector search, real-time insight streaming, report generation, and multi-provider LLM integration, delivering a modular, extensible, and production-ready AI analytics platform.

AI Career Advisor | LLMs, Flask 🏠 GitHub

- Built an LLM-powered career guidance platform that analyzes resumes, identifies skill gaps, and generates personalized learning pathways and role-specific recommendations.
- Developed backend services for resume processing and AI-driven recommendation workflows, enabling scalable and automated career analysis.

EDUCATION

Bachelor of Technology – Computer Science & Engineering, *KL University* 2022 – 2026
CGPA: 9.59 Guntur, India

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02)	2025
Oracle Generative AI Professional	2025
Oracle AI Foundations Associate	2025
MongoDB Certified Developer Associate (C100DEV)	2026

ACHIEVEMENTS

- Ranked among the **Top 10 Finalists** in the **TechGig Coding Contest 2024**, demonstrating strong problem-solving and algorithmic skills.
- Earned **Gold-tier badges** on **HackerRank** and **CodeChef** for proficiency in Data Structures, Algorithms, and Competitive Programming.